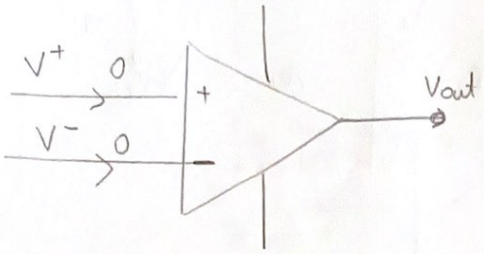
DEEA: laborator 15.12.2022

A.O. IDEAL

1.

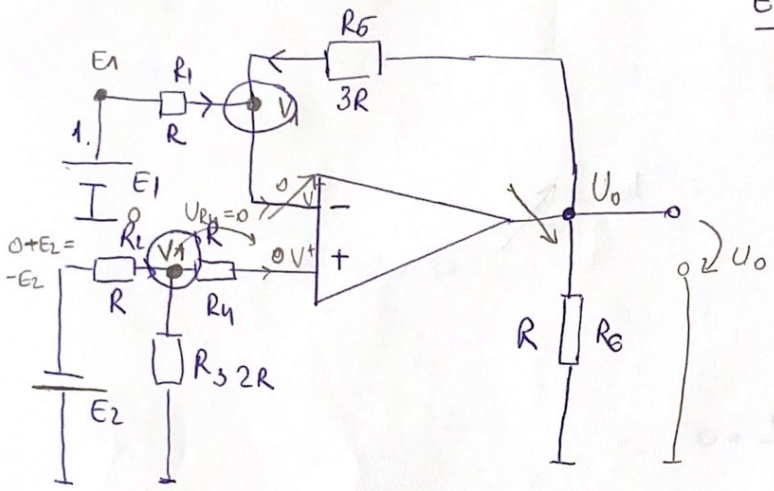


$V_{out} = A(V^+ - V^-)$

$A \to \infty$

$I^+ = I^- = 0$

R.N. $\Rightarrow V^+ = V^-$



$\frac{E_1 - V_1}{R_1} + \frac{U_0 - V_1}{R_5} = 0$

$\frac{E_1 - V_1}{R} + \frac{U_0 - V_1}{3R} = 0$

$\frac{3E_1 - 4V_1 + U_0}{3R} = 0$

$3E_1 - 4V_1 + U_0 = 0$

$\frac{-E_2 - V_1}{R} + \frac{0 - V_1}{2R} = 0$

$-E_2 - 2V_1 = 0 \Rightarrow V_1 = \frac{-E_2}{2}$

$\Rightarrow 3E_1 - 2E_2 + U_0 = 0$

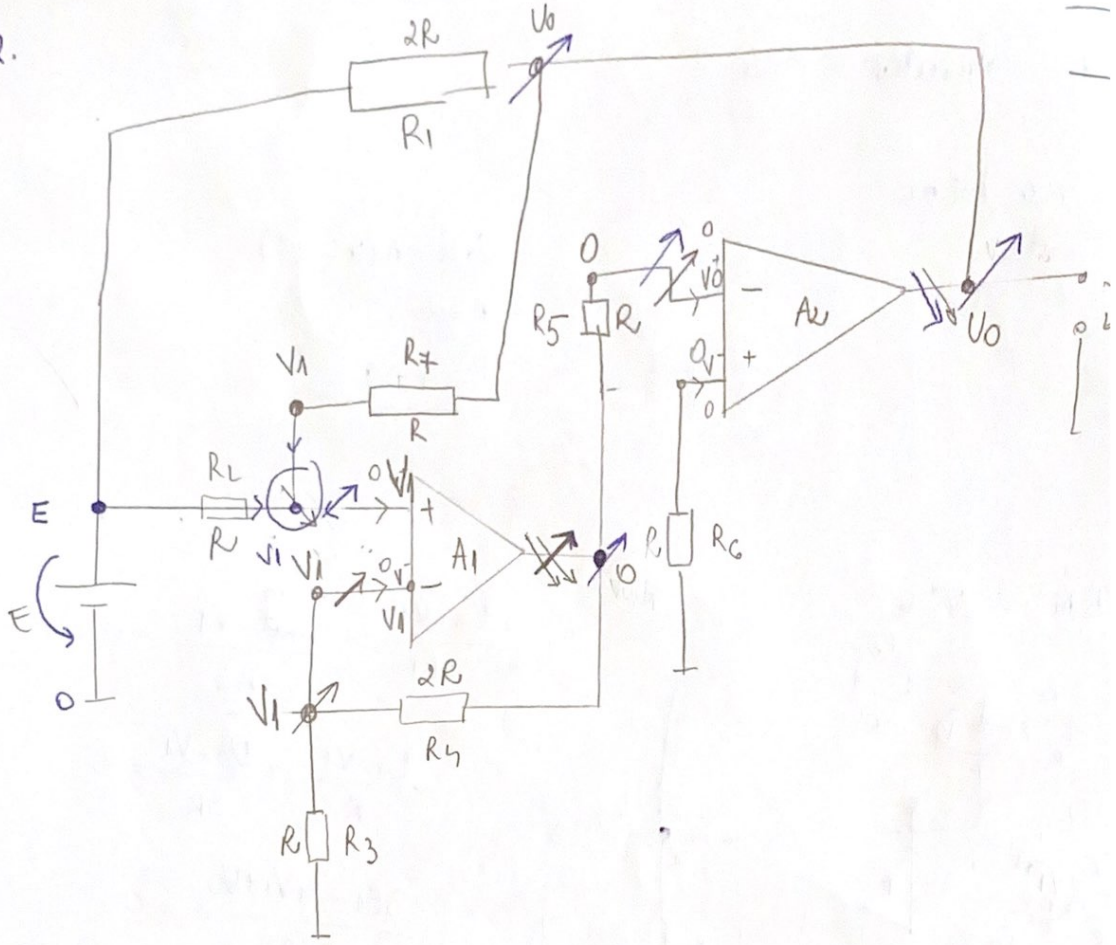
$\Rightarrow U_0 = 2E_2 - 3E_1$

! { Metoda potențialelor la noduri
{ necunoscute: - potențiale
{ ecuații: - KCL (KI)

→ Verificăm R.N.

R.N. $\Rightarrow V^+ = V^-$

2.



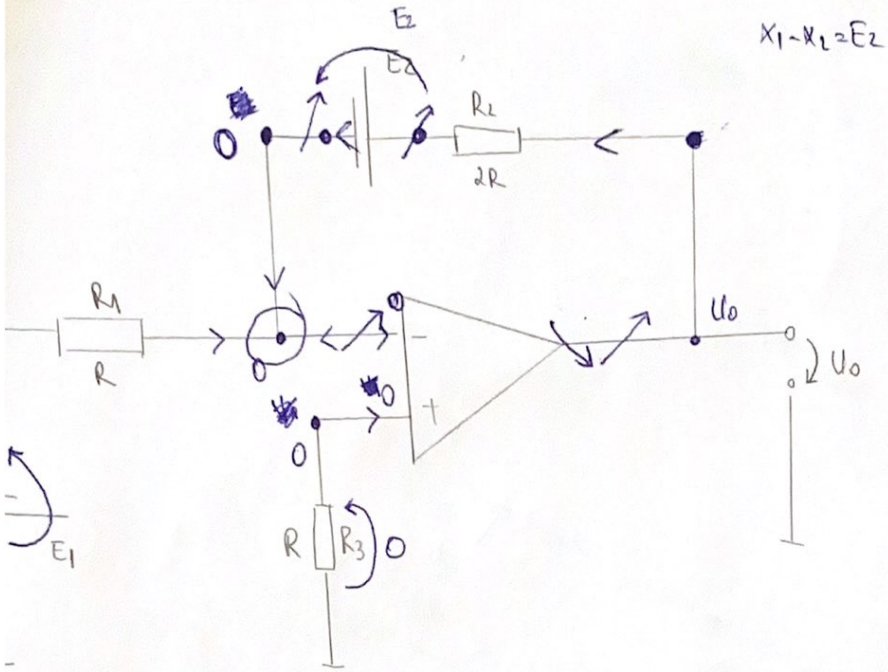
$$0 + \frac{0-V_1}{R_3} + \frac{0-V_1}{R_4} = 0$$

$$\frac{-V_1}{R} - \frac{V_1}{2R} = \frac{-2V_1 - V_1}{2R} = -\frac{3}{2}\frac{V_1}{R} = 0 \Rightarrow V_1 = 0$$

$$\frac{E-V_1}{R_2} + \frac{U_0-V_1}{R_7} + 0 = 0$$

$$\frac{E-V_1}{R} + \frac{U_0-V_1}{2R} = 0$$

$*E + U_0 = 0 \Rightarrow U_0 = -E.$



$X_1 - X_2 = E_2$

~~$-\frac{E_1 - 0}{R_1} + \frac{0 - 0}{R_2} + 0 = 0$~~

~~$-\frac{E_1}{R} =$~~

$$\frac{-E_1 - 0}{R_1} + \frac{U_0 - E_2}{R_2} = 0$$

$$\frac{-E_1}{R} + \frac{U_0 - E_2}{2R} = 0 \rightarrow 2E_1 + E_2 = U_0.$$